

4_Tuples

September 15, 2026

```
[3]: mytuple=(1,2,3,4,5)
      print(mytuple)
      #Hold multiple data types in a tuple
      mytuple1=(1,2.5,3+2j,"Hanu",[9,5.5,6] ,(1,2,3))
      print(mytuple1)
```

```
(1, 2, 3, 4, 5)
(1, 2.5, (3+2j), 'Hanu', [9, 5.5, 6], (1, 2, 3))
```

```
[5]: print(mytuple1[0])
      print(mytuple1[3])
      print(mytuple1[4][1])
```

```
1
Hanu
5.5
```

```
[11]: print(mytuple1[4])
        print(mytuple1[5])
        print(type(mytuple1[4]))
        print(type(mytuple1[5]))
        print(type(mytuple1[3]))
```

```
[9, 5.5, 6]
(1, 2, 3)
<class 'list'>
<class 'tuple'>
<class 'str'>
```

```
[13]: Mlist=[1,2,3,4,5]
        Mtuple=(1,2,3,4,5)
        Mlist[0]=100
        print(Mlist)
        Mtuple[0]=100
        print(Mtuple)
```

```
[100, 2, 3, 4, 5]
```

TypeError

Traceback (most recent call last)

```

/tmp/ipykernel_2432/4053957814.py in <cell line: 0>()
      3 Mlist[0]=100
      4 print(Mlist)
----> 5 Mtuple[0]=100
      6 print(Mtuple)

```

TypeError: 'tuple' object does not support item assignment

```

[14]: #unpacking a tuple
      x, y, z, a, b = mytuple
      print(x)
      print(y)
      print(z)
      print(a)
      print(b)

```

1
2
3
4
5

```

[15]: new_tuple=(26,)
      print(type(new_tuple))

```

<class 'tuple'>

```

[18]: new_tuple1=(10,20,10,10,20,30,40,30,50,60,70,80,90,100)
      print(new_tuple1)
      print(len(new_tuple1)) #prints the length of the tuple
      print(new_tuple1.count(10)) #prints the number of occurrences of 10 in the tuple
      print(new_tuple1.index(20)) #prints the index of the first occurrence of 20 in
      ↪ the tuple
      print(new_tuple1.index(30)) #prints the index of the first occurrence of 30 in
      ↪ the tuple
      print(new_tuple1.index(30, 4)) #prints the index of the first occurrence of 30
      ↪ in the tuple starting from index 4
      print(new_tuple1.index(30, 4, 10)) #prints the index of the first occurrence of
      ↪ 30 in the tuple starting from index 4 and ending at index 10
      print(new_tuple1.index("Hanumanth")) #gives error because "Hanumanth" is not
      ↪ present in the tuple

```

(10, 20, 10, 10, 20, 30, 40, 30, 50, 60, 70, 80, 90, 100)
14
3
1
5
5

```

-----
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_2432/832429188.py in <cell line: 0>()
      7 print(new_tuple1.index(30, 4)) #prints the index of the first occurrence_
    ↪ of 30 in the tuple starting from index 4
      8 print(new_tuple1.index(30, 4, 10)) #prints the index of the first_
    ↪ occurrence of 30 in the tuple starting from index 4 and ending at index 10
----> 9 print(new_tuple1.index("Hanumanth")) #gives error because "Hanumanth" is_
    ↪ not present in the tuple

ValueError: tuple.index(x): x not in tuple

```

```

[19]: # Tuple concatenation
tuple1 = (1, 2, 3)
tuple2 = (4, 5, 6)
tuple3 = tuple1 + tuple2
print(tuple3) # Output: (1, 2, 3, 4, 5, 6)

```

(1, 2, 3, 4, 5, 6)

```

[20]: fruits = ("apple", "banana", "cherry", "date", "elderberry")
sub_fruits = fruits[1:4] # Slicing from index 1 to 3 (4 is not included)
print(sub_fruits) # Output: ('banana', 'cherry', 'date')

```

('banana', 'cherry', 'date')

```

[21]: tuple1=('a','b','c','d','e','f','g')
print(tuple1)
print(tuple1[-3]) #prints the third last element of the tuple
print(tuple1[2:5]) #prints the elements from index 2 to 4
print(tuple1[2:]) #prints the elements from index 2 to the end
print(tuple1[:5]) #prints the elements from the start to index 4

```

('a', 'b', 'c', 'd', 'e', 'f', 'g')

e

('c', 'd', 'e')

('c', 'd', 'e', 'f', 'g')

('a', 'b', 'c', 'd', 'e')